



Tools to Support Systematic Reviews in Software Engineering: A Feature Analysis

Christopher Marshall
School of Computing and
Mathematics
Keele University
Staffordshire UK
c.marshall@keele.ac.uk

Pearl Brereton
School of Computing and
Mathematics
Keele University
Staffordshire UK
o.p.brereton@keele.ac.uk

Barbara Kitchenham
School of Computing and
Mathematics
Keele University
Staffordshire UK
b.a.kitchenham@keele.ac.uk

ABSTRACT

Background The labour intensive and error prone nature of the systematic review process has led to the development and use of a range of tools to provide automated support.

Aim The aim of this research is to evaluate a set of candidate tools that provide support for the overall systematic review process.

Method A feature analysis is performed to compare and evaluate four candidate tools.

Results Each of the candidates has some strengths and some weaknesses. SLuRp has the highest overall score and SLRTOOL has the lowest overall score. SLuRp scores well on process management features such as support for multiple users and document management and less well on ease of installation.

Conclusions Although the tools do not yet support the whole systematic review process they provide a good basis for further development. We suggest a community effort to establish a set of features that can inform future tool development.

Categories and Subject Descriptors

D.2.m [Software Engineering]: Miscellaneous

General Terms

Measurement, Documentation, Performance, Design

Keywords

Systematic review, systematic literature review, feature analysis, systematic review support tools

1. INTRODUCTION

Evidence-based Software Engineering (EBSE) aims to “provide a means by which current best evidence from research can be integrated with practical experience and human values in the decision making process regarding the development and maintenance of software” [1]. Since EBSE was defined in 2004, there has been a wealth of contributions from software engineering (SE) researchers, many of whom have employed the

systematic (literature) review methodology as an integral part of their work [2].

Systematic reviews (SRs) are concerned with rigorously evaluating empirical evidence from relevant literature, in an attempt to address a particular issue or topic of interest [3]. SRs differ from conventional literature reviews in being formally planned and methodically executed [4]. An SR comprises several discrete stages that can be grouped into three phases; namely, the planning phase, the conduct phase and the reporting phase (see Figure 1).

Despite having a well-defined process, SRs are quite labour intensive and error prone [5,6,7,8] making them prime candidates for automated support. A number of tools to support the SR process were identified through a recent mapping study which found a predominance of visualisation and text mining techniques to support study selection, data extraction and data synthesis [9]. The mapping study identified three tools which aim to support SRs across all (or at least many) of the stages of the process.

The study reported here aims to evaluate these three ‘whole process’ tools together with a further known (but unpublished) system which also aims to support systematic reviewers in SE. The study takes the form of a feature analysis and is the first step toward the development of a rigorous evaluation framework for tools that support SRs. A set of features that such tools should possess is proposed and each tool is scored against each feature. The strengths and weaknesses of each tool, in terms of how well it provides each of the features, are discussed.

The paper is organised as follows. Section Two describes the DESMET method and, in particular, the feature analysis approach. Section Three introduces the candidate tools and describes the set of features used as the basis for the evaluation, the scoring process and the method used to calculate an overall score for each of the candidate tools. Section Four presents the results of the feature analysis, followed by a discussion in Section Five. Section Six presents some conclusions drawn from the study and implications for future work.

2. METHOD

This section describes the approach taken to evaluate the four candidate tools. An overview of the DESMET methodology is followed by a description of the chosen evaluation method, i.e. the feature analysis approach.

2.1 DESMET

DESMET is a methodology for evaluating methods or tools. It defines nine different evaluation types and a set of criteria to

assist the evaluator in selecting the most appropriate one based on their needs [10]. A DESMET evaluation is context-dependent and comparative. This means it is not used to rank tools in terms of effectiveness, but instead to retrieve information on which to base a decision about a tool's suitability in a particular context [10]. The context in this case is an academic one, specifically where researchers are undertaking an SR within the SE domain.

When adopting the DESMET methodology, the first stage is to select an evaluation type. The types of evaluation available can be categorised according to the aspects of the tool that are to be examined. If the primary aspects of a tool to be evaluated are the effect it has within an organisation, then quantitative methods of evaluation are deemed most appropriate. If, however, the objective of the evaluation is more concerned with the suitability of a tool in a given setting, then this can be better determined using a qualitative form of evaluation. Both categories of evaluation can be organised as a formal experiment, case study or survey. Qualitative forms of evaluation, however, can also be organised as a feature analysis. Due to the context of this study, a qualitative form of evaluation has been selected.

2.2 Feature Analysis

Feature analysis is a qualitative form of evaluation involving the subjective assessment of the relative importance of different features plus an assessment of how well each of the features is implemented by the candidate tools [11]. It is an established evaluation method in SE [13, 14]. The feature sets are based on the requirements that users have for the particular tasks that they expect the tool to support [11, 12]. For this study, a feature analysis is organised as an initial screening and focuses on evaluating simple features. Simple features relate to aspects that are either present, partially present or absent [10].

3. CANDIDATES, FEATURES AND SCORING

This section provides an overview of the four candidate tools, the feature set against which the tools are evaluated and the approach taken to scoring the candidates. As far as we are aware, the candidate tools are the only ones to date that aim to address the overall SR process in SE.

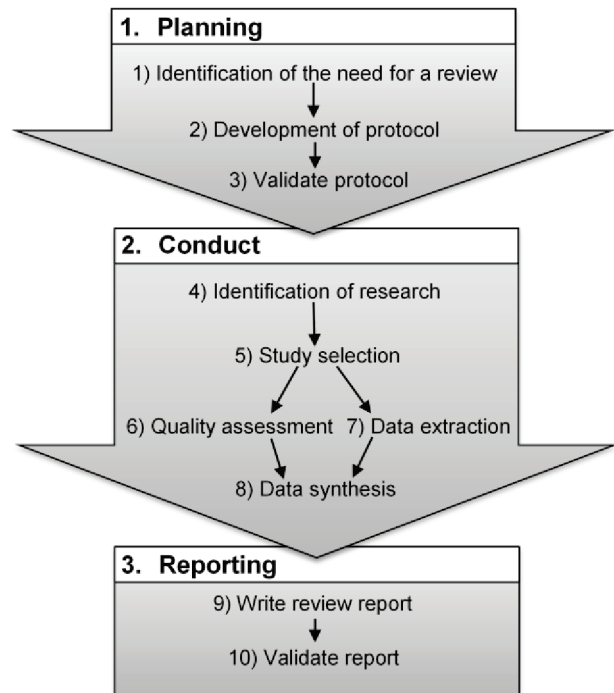
3.1 Candidate Tools

The four candidate tools are:

- a) **Systematic Literature unified Review Program (SLuRp)** which is described as an open source web-enabled database that supports the management of SRs [15]. The tool has been developed using Java and SQL.
- b) **State of the Art through systematic review (StArt)** which aims to provide support for each stage of the SR process in SE [16].
- c) **SLR-Tool**, developed in Java, which is described as a freely-available tool to support each stage of the SR process in SE [17].
- d) **SLRTOOL**¹ which aims to support the SR process in SE, amongst other disciplines. The developers state that the guidelines, established by Kitchenham and Charters, underpin its design. SLRTOOL was not identified by the mapping study reported in [9].

At the beginning of the study, the developers of each tool were contacted and informed that their tool had been selected as a

Figure 1. 10-Stage Systematic Review Process



candidate for the feature analysis. They were asked if they could provide the most up-to-date version of their tool plus any relevant literature or documentation that supports it. A developer of SLuRp, based at University of Hertfordshire, invited the evaluation team to attend a demonstration of the tool. The team that developed StArt provided a recently updated version of the tool, a related publication and a link to a video tutorial. The developers of SLR-Tool provided an updated version of their tool plus a user manual and installation guide. A developer of SLRTOOL responded and informed the author that his query had been forwarded to a more suitable member of the team. Following this initial interaction, no further response was received.

3.2 Set of Features

As well as covering technical aspects, features should also include economic, cultural and quality aspects [10]. A feature can be decomposed into subfeatures and further broken down into subsubfeatures if required. Features for this study are based on:

- the experiences of performing SRs reported in the literature [4, 5, 6, 7],
- a preliminary screening of the four candidate tools,
- discussions between the authors.

The features are divided into four sets relating to economics, ease of introduction, SR activity support and process management (see Table 1). The following subsections describe the features in each of these sets.

3.2.1 Feature Set 1: Economic

This set concerns economic factors relating to the initial cost of the tool and the subsequent support for maintaining (or upgrading) the tool. For this study, highest scores are awarded if no initial payment is required (F1-SF01) and the tool is well (and freely) maintained by its developers, including having regular updates and a single point of contact for users to obtain support if needed (F1-SF02).

¹ www.slrtool.org

Table 1. Features used in the analysis

<i>id</i>	<i>Feature Set</i>	<i>id</i>	<i>Subfeature</i>	<i>Subfeature Level of Importance</i>	<i>Interpretation of Judgement Scale</i>	<i>Feature set Importance Weighting</i>
F1	Economic	F1-SF01	The tool does not require financial payment to use.	HD	J11	0.1
		F1-SF02	Maintenance	HD	J11	
F2	Ease of introduction and setup	F2-SF01	The tool has reasonable system requirements.	M	J11	0.2
		F2-SF02	Simple installation and setup.	HD	J12	
		F2-SF03	There is an installation guide.	HD	J11	
		F2-SF04	There is a tutorial.	HD	J11	
		F2-SF05	The tool is self-contained.	HD	J11	
F3	SR activity support	F3-SF01	Protocol development	D	J13	0.4
		F3-SF02	Protocol validation	D	J13	
		F3-SF03	Supports automated searches	HD	J13	
		F3-SF04	Study selection and validation	HD	J13	
		F3-SF05	Quality assessment and validation	HD	J13	
		F3-SF06	Data extraction and validation	HD	J13	
		F3-SF07	Automated analysis	HD	J13	
		F3-SF08	Text analysis	N	J11	
		F3-SF09	Meta-analysis	N	J11	
		F3-SF10	Report write up	N	J13	
		F3-SF11	Report validation	N	J13	
F4	Process Management	F4-SF01	Support for multiple users	M	J11	0.3
		F4-SF02	Document management	M	J11	
		F4-SF03	Security	D	J11	
		F4-SF04	Management of roles	HD	J11	
		F4-SF05	Support for multiple projects	M	J11	

3.2.2 Feature Set 2: Ease of introduction and setup

This feature set focuses on the level of difficulty inherent in setting up and using the tool for the first time. Each tool should:

- have reasonable system requirements (F2-SF01) and not require any advanced hardware or software to function,
- have a simple installation and setup procedure (F2-SF02) that is supported by an installation guide (F2-SF03) and/or a tutorial (F2-SF04),
- be as self-contained as possible i.e. able to function, primarily, as a stand-alone application with minimal requirements for other external technologies (F2-SF05).

3.2.3 Feature Set 3: SR Activity Support

These features relate to how well the tool supports each of the three main phases of an SR and the steps within these phases.

Planning Phase

For this phase, the tool should support the collaborative development of a review protocol, using a template, and the control of versions, to keep track of any changes to the protocol during its development (F3-SF01). It should also support validation of the protocol (F3-SF02). This might be achieved by enabling evaluation checklists to be distributed to and completed by members of a review team.

Conduct Phase

For the conduct phase, the tool should support:

- **automated searching** for relevant papers (F3-SF03). Ideally, the user should be able to perform an

automated search from within the tool which should identify duplicate papers and handle them accordingly.

- **study selection and validation** (F3-SF04). In particular, the tool should provide support for a multi-stage selection process (i.e. title/abstract then full paper), for multiple users to apply the inclusion/exclusion criteria independently and a facility to reconcile disagreements.
- **quality assessment and validation** (F3-SF05). The tool should enable the use of suitable quality assessment criteria, should allow multiple users to perform the scoring and should provide a facility to resolve conflicts.
- **data extraction** (F3-SF06). In particular, the tool should support the extraction and storage of qualitative data using classification and mapping techniques. In addition, the extraction of quantitative data, which manages specific numerical information from a reported study, should also be supported.
- **data synthesis** (F3-SF07). The tool should be able to provide automated analysis of extracted data. Other types of analysis, such as text analysis (F3-SF08) and meta-analysis (F3-SF09), would also be useful.

Reporting Phase

The tool should support the reporting phase of the SR process. This might be achieved using a template to assist the write-up (F3-SF10) and using automated checklists to support the validation (F3-SF11).

Table 2. JI1 Interpretation of Judgement Scale

<i>Is the feature present?</i>	<i>Score</i>
Yes	1
Partly	0.5
No	0

3.2.4 Feature Set 4: Process Management

This set of features relates to the management of an SR. Undertaking an SR is a collaborative process. Therefore, the tool should allow multiple users to work on a single review (F4-SF01). It should support document management (F4-SF02), in particular, managing large collections of papers, studies and the relationships between them. The tool should be secure (F4-SF03) and include a user log-in or similar system. It should be able to manage the roles of users (F4-SF04). For example, it would be useful to state which users will perform certain activities (e.g. study selection, quality assessment, data extraction etc.) and allocate papers accordingly. Finally, the tool should be able to support multiple SR projects (F4-SF05).

3.3 Scoring Candidate Tools

In this section we look at three elements of the scoring process:

- scoring each tool against each feature to produce a raw score,
- assigning a level of importance to each feature which is used as a weighting (i.e. a multiplier) to convert raw scores to weighted scores for each feature,
- determining scores for each feature set and an overall score for each candidate tool.

Each tool was initially scored against each feature by the first author (CM). The scores were then discussed by all of the authors to produce a set of validated raw scores. A spreadsheet was used to record raw scores, weighted scores and overall scores.

The following sections provide some more details about the judgement scale and its interpretation for specific features, the assignment of a level of importance to each feature and the approach taken to calculating overall scores for each of the tools.

3.3.1 Judgement Scale and its Interpretation

A single simple judgement scale was used to score the features. Where a feature is fully present or strongly supported it was awarded a score of 1, where it was partly present or partially supported it was awarded a score of 0.5 and where it was absent or minimally supported it was awarded a score of 0.

The judgement scale was interpreted for each of the features in one of three ways, labeled JI1, JI2 and JI3 in Table 1. The interpretations are shown in Tables 2, 3 and 4.

Table 4. JI3 - Interpretation of Judgement Scale

<i>Is the activity supported?</i>	<i>Score</i>
Yes - Fully	1
Partly Support is limited. Some aspects of the activity are not supported.	0.5
No	0

Table 3. JI2 - Interpretation of Judgement Scale

<i>Is the tool simple to install and setup?</i>	<i>Score</i>
Yes	1
Some difficulties The tool could be installed, but there were a number of slight difficulties throughout the process.	0.5
No The tool could be installed but the process was very difficult.	0
No - The tool could not be installed.	

3.3.2 Level of Importance

An effective tool is one that includes features that are most important to its target users [10]. Kitchenham et al. state that if a tool fails to include a mandatory feature, then it is, by definition, unacceptable [10]. Non-mandatory features allow the evaluator to judge the relative merit of a group of otherwise acceptable tools [10]. For this study, a feature can be considered Mandatory (M) or one of three gradations of desirability; namely, Highly Desirable (HD), Desirable (D) or Nice to have (N). Table 5 shows the multiplier (i.e. the weighting) associated with each level of importance. The level of importance assigned to each feature set is shown in Table 1. The importance levels were determined through discussion between the authors.

3.3.3 Feature Set and Overall Scores

As indicated above, a weighted score for each feature is calculated by multiplying the raw score by the importance weighting for that feature. These weighted scores can be combined to determine a percentage score for each feature set (as shown for example in column 5 of Table 7).

The percentage score for a feature set is determined as follows:

$$\text{Percentage Score} = \frac{\text{Sum of Weighted Scores}}{\text{Maximum Score}} \times 100\%$$

The maximum score for a feature set is assumed to be the sum of the weighted scores where all features in the set are fully present (or fully supported).

For example, Feature Set 1 (F1) has two subfeatures (F1-SF01 and F1-SF02). F1-SF01 has been classified as highly desirable (HD). This means the maximum weighted score for this subfeature is three. Similarly, F1-SF02 has been classified as HD so its maximum score is also three. Therefore, the maximum score for F1 is six. Similarly, the maximum scores for the remaining feature sets are 16 for Feature Set 2, 23 for Feature Set 3 and 17 for Feature Set 4.

An overall percentage score for each tool can be determined by taking a (weighted) average of the percentage scores for each feature set. Since there are a different number of subfeatures in each of the feature sets it is necessary to use normalised scores (i.e. the percentage scores) for this.

For this calculation, we use the Feature Set weighting shown in Table 6. The values here emphasise support for SR activities (F3) and for process management (F4). Other weightings could be used, perhaps to emphasise usability, as tools to support SRs become more mature. The overall score for each tool can be determined using the following equation:

$$Overall\ score = \frac{\sum_{i=1}^4 (w_i TP_i)}{\sum_{i=1}^4 (w_i)} \quad (3.1)$$

where w_i is the weighting for the i^{th} feature set and TP_i is the percentage score for the i^{th} feature set.

This study was intended to assess the potential of SR support tools from the viewpoint of the tasks that are undertaken during a collaborative SR. For this reason, we chose values of the overall weights that emphasised the feature sets that provide the functions needed by an SR research team performing an SR (i.e. Feature Sets 3 and 4) and reduced the weights for the feature sets related to economic and installation issues (i.e. Feature Sets 1 and 2) that are generic tool issues.

4. RESULTS

This section reports the post-validation scores, indicating which of these were modified by the validation process, and the overall scores for each of the candidate tools. The results for all of the candidate tools are summarised in Table 11.

4.1 Results for SLuRp

Table 7 presents the scores for SLuRp.

4.1.1 Feature Set 1

SLuRp is free to use and can be accessed from the development team's website [15]. The tool is well maintained, regularly updated and provides a single point of contact for a user to obtain help if needed. SLuRp scored full marks for this feature set.

4.1.2 Feature Set 2

SLuRp can be used at the developer's website, on request. However, it is likely most users will opt to download, install and implement the tool locally. SLuRp has a complex setup. To configure a full version of SLuRp, a number of external technologies must also be installed; namely, Tomcat, MySQL, LaTeX and R. SLuRp can be used without installing LaTeX and R, but doing so will remove some of its features. Some installation instructions can be found at the tool's website. Currently, there is no tutorial. SLuRp scored 6.5 out of 16 marks for this feature set.

4.1.3 Feature Set 3

To-date, SLuRp does not support protocol development or automated searches. Most other stages, however, are supported by the tool. To assist quality assessment and study selection, SLuRp allows users to, independently, define and apply multiple criteria, throughout a multi-stage selection process. In addition, SLuRp identifies disagreements between quality scores, inclusions and exclusions. To resolve disputes, SLuRp supports moderation whereby a user, outside of the conflict, acts as a mediator. To assist data extraction, SLuRp allows users to design and apply two types of data extraction form; namely, a coding form and a

Table 5. Level of Importance of a feature

<i>Importance</i>	<i>Multiplier</i>
Mandatory (M)	*4
Highly Desirable (HD)	*3
Desirable (D)	*2
Nice to have (N)	*1

Table 6. Feature Set Weighting

<i>Feature Set</i>	<i>Weight</i>
F1	0.1
F2	0.2
F3	0.4
F4	0.3

performance form. The coding form allows the user to extract and record qualitative data about each paper. It is particularly useful for classification and mappings. The performance form allows users to extract more specific quantitative data from a study. In the supporting paper, a number of features that support automated analysis are described [15]. Unfortunately, during the evaluation, these features were not able to be fully-tested. These features have, however, been observed during a demonstration of the tool by one of its lead developers. SLuRp provides facilities for text analysis using an embedded SQL editor. In addition, meta-analysis is supported (providing R has been installed). The final report can be written (in full) within the tool providing LaTeX is installed. In total, SLuRp scored 10 out of 23 marks for this feature set.

Table 7. Scores for SLuRp

<i>Feature Set</i>	<i>Sub Feature</i>	<i>Weighted Score</i>	<i>Feature Set Score</i>	<i>% Feature Set Score</i>
F1	F1-SF01	3	6/6	100%
	F1-SF02	3		
F2	F2-SF01	2	6.5/16	41%
	F2-SF02	0		
	F2-SF03	3		
	F2-SF04	0		
	F2-SF05	1.5		
F3	F3-SF01	0	10/23	43%
	F3-SF02	0		
	F3-SF03	0		
	F3-SF04	1.5		
	F3-SF05	3		
	F3-SF06	1.5		
	F3-SF07	1.5		
	F3-SF08	1		
	F3-SF09	0.5		
	F3-SF10	1		
	F3-SF11	0		
F4	F4-SF01	4	17/17	100%
	F4-SF02	4		
	F4-SF03	2		
	F4-SF04	3		
	F4-SF05	4		
Total Score			Overall % Score Using Feature Set Weightings	
39.5/62			65.4%	

4.1.4 Feature Set 4

SLuRp allows multiple users to work on a single review and allows multiple projects to be undertaken. The tool contains a number of useful document management features. Papers can be imported into the tool using BibTeX. As part of the import process, SLuRp will attempt to attach a full copy of a paper, automatically. If an attachment fails, a link to its location is provided. SLuRp assists with management of roles. The “super-user” can manage various levels of access for other users and assign them to undertake particular activities. SLuRp implements a secure login system, which requires a username and password on each visit. SLuRp scored full marks for this feature set.

4.1.5 Modifications of scores

Two scores were modified as a result of the validation process. Originally, full marks were awarded for SLuRp’s support of data extraction. However, during the validation process, it became clear that the lead evaluator (CM) failed to fully understand the “performance form” and how to use it effectively. Although demonstrated by one of SLuRp’s developers, it was agreed that full marks could not be justified until this feature had been properly tested. As a result, the initial score was reduced. In addition, the original score allocated for SLuRp’s support of quality assessment, has been modified. Initially, it was considered that the tool only provided partial support for this stage. However, it was decided that SLuRp supported this activity better than first thought. In particular, allowing multiple users to apply criteria independently and resolve conflicts through moderation, helped to increase its score.

4.1.6 Overall Score

As indicated in Section 3.3.3, the overall score for the tool is calculated using Equation 3.1 and the Feature Set Weightings shown in Table 6. For SLuRp, the overall score is 65.4%.

4.2 Results for SLRTOOL

Table 8 presents the scores for SLRTOOL.

4.2.1 Feature Set 1

SLRTOOL can be accessed from the developer’s website, free of charge. The tool, however, does not seem to be well maintained. During this study, the tool’s website was not always available and, although new features have been planned, the tool has not been updated. SLRTOOL scored 3 out of 6 marks for this feature set.

4.2.2 Feature Set 2

Following a registration process, the tool can be used online, at the developer’s website. Alternatively, the source-code and database script can be downloaded (from the same website) for local installation. This setup requires an Apache web server, PHP and MySQL database. The installation process, although not entirely straight forward, was considered reasonable. Brief installation instructions were found at the website. Currently, there is no tutorial. SLRTOOL scored 11.5 out of 16 for this feature set.

4.2.3 Feature Set 3

SLRTOOL does not support protocol development or validation. A facility, which allows the user to perform an internal, automated search, has been developed. However, whilst initially promising, the feature is rather limited and only allows informal, ad-hoc keyword searches of Google Scholar. Whilst there is potential, in its current state this feature does not provide adequate support. SLRTOOL aims to support study selection. However, support is limited. In particular, users are unable to apply

Table 8. Scores for SLRTOOL

Feature Set	Sub Feature	Weighted Score	Feature Set Score	% Feature Set Score
F1	F1-SF01	3	3/6	50%
	F1-SF02	0		
F2	F2-SF01	4	11.5/16	72%
	F2-SF02	1.5		
	F2-SF03	3		
	F2-SF04	0		
	F2-SF05	3		
F3	F3-SF01	0	4.5/23	20%
	F3-SF02	0		
	F3-SF03	0		
	F3-SF04	0		
	F3-SF05	1.5		
	F3-SF06	1.5		
	F3-SF07	1.5		
	F3-SF08	0		
	F3-SF09	0		
	F3-SF10	0		
	F3-SF11	0		
F4	F4-SF01	2	10/17	59%
	F4-SF02	2		
	F4-SF03	2		
	F4-SF04	0		
	F4-SF05	4		
Total Score			Overall % Score Using Feature Set Weightings	
29/62			45.1%	

inclusion/exclusion criteria using a multi-stage process and multiple users cannot perform their selections independently. Quality assessment is partially supported. A quality criterion, using a simple, nominal scale, can be applied to studies. Data extraction is partially supported by the tool. Users can design a three-tier classification form to assist with extraction. Analysis of the data is, however, limited. SLRTOOL can perform analysis on certain aspects of the review; such as, study selection, quality assessment, publisher and year of publication. The tool produces a number of charts to visualise these findings. However, automated analysis of extracted data is not performed by the tool. SLRTOOL scored 4.5 out of 23 for this feature set.

4.2.4 Feature Set 4

The tool partially supports multiple users. Once a project has been created, new users, labelled “collaborators”, can be added. Providing a user has been registered, they can be located and added to a project using the tool’s user-search facility. Each user can be a “collaborator” on multiple projects and, at the same time, the “lead-user” of their own projects. SLRTOOL does not, however, support the management of roles within a project. Support for document management is also limited. Although papers can be exported from the tool as a BibTeX file, they cannot be imported in bulk, using the same method. Papers have to be

manually imported one at a time. Once papers are stored, however, the tool provides reasonable facilities to manage and organise them. SLRTOOL requires each user to register a username and password, which must be entered at each visit. SLRTOOL scored 13.5 out of 17 for this feature set.

4.2.5 Modifications of scores

Four scores were modified as a result of the validation process. Initially, partial marks were awarded for SLRTOOL's support of an automated search. However, this score was reduced. It was agreed that, although there is potential, the tool does not provide enough support to fulfill the rigour required of an SRs search process. The score for SLRTOOL's support of study selection was also modified. Initially, partial marks were awarded for its support of the activity. However, once discussed, it was agreed that support was too limited. In particular, users were unable to perform a multi-stage selection process, independently. As a result, the score was reduced. Finally, the initial scores received for SLRTOOL's support of multiple users and management of roles, have been revised. The foundations for collaboration, amongst multiple users, are in place. Users can be easily located, added and removed from a project, at any time. However, the tool's support for what are considered the collaborative aspects of an SR is, generally, quite limited. Due to this, both scores were reduced.

4.2.6 Overall Score

Using Equation 3.1 and the Feature Set Weightings shown in Table 6, the overall score for SLRTOOL is 45.1%.

4.3 Results for StArt

Table 9 presents the scores for StArt.

4.3.1 Feature Set 1

StArt is free to use and can be downloaded from the developer's website [16]. The tool is well maintained and regularly updated with new features and fixes. In addition, there exists a single point of contact for user assistance. StArt scored full marks for this feature set.

4.3.2 Feature Set 2

StArt must be downloaded from the developer's website and installed locally. The tool's setup was simple and easy to perform, using a full installation wizard. To assist users, the developers have created an introductory video, providing an overview of the tool and its key features. StArt is entirely self-contained and does not require any external applications to be installed. StArt scored 14.5 out of 16 for this feature set.

4.3.3 Feature Set 3

StArt provides a reasonably detailed template for user's to develop a protocol. Its validation, however, is not supported. StArt cannot apply search strings directly to digital libraries and retrieve papers automatically. The developers claim this limitation is due to security rules imposed on the search engines [15]. However, StArt allows searches to be managed using "search sessions". For each search, a user defines a new "search session". Each "search session" corresponds to a particular resource (that is to be searched) and a search string. Once the user has performed the search, its results are imported and stored, within the "search session". StArt provides support for a two-stage study selection process. Quality assessment, however, is not supported by the tool. StArt provides partial support for data extraction. Classification forms, designed using the protocol template, can be used to assist this stage. StArt will perform analysis on extracted

Table 9. Scores for StArt

<i>Feature Set</i>	<i>Sub Feature</i>	<i>Weighted Score</i>	<i>Feature Set Score</i>	<i>% Feature Set Score</i>
F1	F1-SF01	3	6/6	100%
	F1-SF02	3		
F2	F2-SF01	4	14.5/16	90%
	F2-SF02	3		
	F2-SF03	3		
	F2-SF04	1.5		
	F2-SF05	3		
F3	F3-SF01	1	8.5/23	37%
	F3-SF02	0		
	F3-SF03	1.5		
	F3-SF04	1.5		
	F3-SF05	0		
	F3-SF06	1.5		
	F3-SF07	1.5		
	F3-SF08	1		
	F3-SF09	0		
	F3-SF10	0.5		
	F3-SF11	0		
F4	F4-SF01	0	6/17	35%
	F4-SF02	2		
	F4-SF03	0		
	F4-SF04	0		
	F4-SF05	4		
<i>Total Score</i>			Overall % Score Using Feature Set Weightings	
35/62			53.3%	

data. The tool employs a number of visualisations to present this analysis. Analysis for quantitative data is, however, limited. StArt includes an interesting text analysis feature. The tool generates a "score" for each paper. A score is calculated by matching keywords from a paper's title and abstract, with keywords defined in the protocol. In addition, using the same method, the tool calculates a similarity statistic between papers. Meta-analysis, however, is not supported by the tool. StArt provides partial support for reporting the review. Tables and charts, produced by the tool, can be exported for use in a report. In addition, users can export the raw data direct to Excel, for further analysis. StArt scored 8.5 out of 23 for this feature set.

4.3.4 Feature Set 4

StArt does not support multiple users and, therefore, management of roles. Document management is, however, partially supported by the tool. Papers can be imported into the tool in bulk. For this process, StArt supports a variety of reference file formats; including, BibTeX, MEDLINE, RIS and Cochrane. The tool provides useful facilities to manage, sort and organise papers. StArt does not store full papers. Only a paper's location (providing it is locally stored) can be managed by the tool. StArt does not include any features for security. The tool does, however,

allow multiple projects to be undertaken. StArt scored 6 out of 17 for this feature set.

4.3.5 Modifications of scores

Five scores were modified as a result of the validation process. Initially, full marks were awarded for StArt's support of protocol development. During the validation process, however, it was highlighted that StArt does not provide support for version control. Therefore, the score was reduced. The original score awarded for quality assessment, was also modified. It was agreed that, although a "score" (described in Section 4.3.3) is generated by the tool, its calculation process does not reflect the proper procedure for quality assessment in an SR. Therefore, the feature's score was reduced. The initial score for data extraction was also reduced. Initially, full marks were awarded for StArt's support of this stage. During the validation process, however, it was agreed that support is primarily targeted toward mapping studies rather than full SRs. In addition, data extraction is generally considered a collaborative activity. Since StArt does not support multiple users, support for this activity is, therefore, limited. Finally, the initial score awarded for StArt's support of automated analysis, was reduced. Originally, full marks were awarded for this feature. However, it was agreed during the validation process that StArt focuses, primarily, on analysing qualitative data. Support for quantitative data analysis is limited.

4.3.6 Overall Score

Using Equation 3.1 and the Feature Set Weightings shown in Table 6, the overall score for StArt is 53.3%.

4.4 Results for SLR-Tool

Table 10 shows the scores for SLR-Tool.

4.4.1 Feature Set 1

SLR-Tool is free to use and can be downloaded from its developer's website. However, the website remains (to-date) in poor condition and it seems that the tool has not been updated for some time. SLR-Tool scored 4.5 out of 6 for this feature set.

4.4.2 Feature Set 2

SLR-Tool requires an installation of MySQL to function. This component is relied on heavily by the tool. Its setup procedure is supported with a reasonably effective installation manual. During the installation process an option is available to load an example SR project into the tool. When combined with the user manual, this serves as an effective tutorial. SLR-Tool scored 14.5 out of 16 for this feature set.

4.4.3 Feature Set 3

SLR-Tool provides a template for users to develop a protocol. The background, justification, research questions, search strategy (including multiple sources and search strings), quality criteria and study selection criteria can all be defined. The protocol's validation, however, is not supported by the tool. In addition, the tool does not provide support for an automated search. SLR-Tool tries to support a multi-stage study selection process, with limited success. Users can apply the study selection criteria, defined in the protocol. Papers can be included/excluded during a "first review" and "second review". In addition, SLR-Tool partially supports quality assessment. During the protocol's development, users design a simple quality assessment questionnaire, which can be applied to included studies. Data extraction is also supported by the tool, all be it, in a limited capacity. Users design classification forms to extract, primarily, qualitative data. SLR-Tool provides effective support for data analysis. The tool generates a variety of

Table 10. Scores for SLR-Tool

Feature Set	Sub Feature	Weighted Score	Feature Set Score	% Feature Set Score
F1	F1-SF01	3	4.5/6	75%
	F1-SF02	1.5		
F2	F2-SF01	4	14.5/16	90%
	F2-SF02	3		
	F2-SF03	1.5		
	F2-SF04	3		
	F2-SF05	3		
F3	F3-SF01	2	10/23	43%
	F3-SF02	0		
	F3-SF03	0		
	F3-SF04	1.5		
	F3-SF05	1.5		
	F3-SF06	1.5		
	F3-SF07	3		
	F3-SF08	0		
	F3-SF09	0		
	F3-SF10	0.5		
	F3-SF11	0		
F4	F4-SF01	0	6/17	33%
	F4-SF02	2		
	F4-SF03	0		
	F4-SF04	0		
	F4-SF05	4		
Total Score			Overall % Score Using Feature Set Weightings	
35/62			53.2%	

charts to visualise findings. Charts can be exported from the tool for use in written reports. SLR-Tool scored 10 out of 23 for this feature set.

4.4.4 Feature Set 4

SLR-Tool does not support multiple users nor, therefore, management of roles. Document management is, however, partially supported. The developers indicate that SLR-Tool is compatible with a range of reference file formats (including, BibTeX, EndNote and RIS) which can be used to import collections of documents [17]. This feature, however, failed to work during our evaluation and we were only able to import papers individually. Once papers are stored, SLR-Tool offers reasonable support for their organisation and management. The tool supports multiple projects to be undertaken. SLR-Tool scored 6 out of 17 for this feature set.

4.4.5 Modifications of scores

Three scores were modified. Initially, full marks were awarded for SLR-Tool's installation guide, however, during validation, it was agreed that its content did not sufficiently cover how to setup the MySQL component. Therefore, the score was reduced. The original score awarded for SLR-Tool's tutorial, was increased. Initially, partial marks were awarded for this feature. However, it

Table 11. Feature Set Scores and Overall Scores

	<i>F1</i> (scores out of 6)		<i>F2</i> (scores out of 16)		<i>F3</i> (scores out of 23)		<i>F4</i> (scores out of 17)		<i>Total</i> (score out of 62)		<i>Overall Score</i> ²
<i>SLuRp</i>	6	100%	6.5	41%	10	43%	17	100%	39.5	64%	65.4%
<i>StArt</i>	6	100%	14.5	90%	8.5	37%	6	35%	35	56%	53.3%
<i>SLR-Tool</i>	4.5	75%	14.5	90%	10	43%	6	35%	35	56%	53.2%
<i>SLRTOOL</i>	3	50%	11.5	72%	4.5	20%	10	59%	29	46%	45.1%

was agreed that the ability to load an example project into the tool, is a highly useful feature (and more useful than initially thought). Finally, partial marks were originally awarded for SLR-Tool's support of role management. The tool allows the user to make note of who (i.e. which members of the review team) will perform certain activities; specifically, the search, study selection and quality assessment. However, since the tool does not support multiple users, it was agreed that management of roles cannot be supported effectively. Therefore, the score was reduced.

4.4.6 Overall Score

Using Equation 3.1 and the Feature Set Weightings shown in Table 6, the overall score for SLR-Tool is 53.2%.

5. DISCUSSION

This section presents a discussion of the results of the feature analysis highlighting the main strengths and weaknesses of each candidate tool. Limitations of the study are also discussed.

5.1 Discussion of Results

SRs in SE usually take one of two forms. The 'standard' form, aims to address specific research questions relating to SE methods or procedures. The alternative form, termed a mapping study, aims to classify the literature on a specific SE topic [18]. For mapping studies, the search strategy is often less stringent than for standard SRs and quality assessment is not usually required. We consider these slightly different requirements within the discussion of results.

As shown in Table 11, SLuRp achieves the highest overall score of 65.4% and so within the constraints of the study can be considered the most suitable tool to support SRs in SE. Its main strengths are:

- Provides full support for a team-based SR process.
- Can be used for standard SRs as well as for mapping studies (good support for quality assessment).
- Actively supported by its developer.

SLuRp's main weaknesses are its complex installation, lack of support for protocol development and difficulties associated with the use of the performance form, as described in Section 4.1.3.

StArt has an overall score of 53.3%. Its main strengths are:

- Active support and maintenance by its developers.
- Its simple setup procedure.

StArt is the only tool that does not rely on the installation of any additional applications in order to function. One of StArt's weaknesses is an absence of support for multiple users. As a consequence, many of the SR stages that are considered collaborative activities are only partially supported by the tool. In addition, it does not support quality assessment. Since we have assigned quality assessment a mandatory level of importance we

suggest that StArt is not yet suitable for standard SRs (as opposed to mapping studies).

SLR-Tool has an overall score of 53.2%. The tool's main strengths are:

- Strong support for developing a review protocol.
- Effective support provided to new users; notably, the ability to load an example project into the tool.
- Effective support for automated analysis.

Its main weakness is its lack of support for multiple users. Also, as indicated in Section 4.4.4, we were unable to import collections of papers. This meant that papers had to be manually imported on a paper-by-paper basis.

SLRTOOL has the lowest overall score of 45.1%. The tool has a number of promising and potential features, yet fails to implement them effectively. In particular, it is clear that support for collaboration, amongst multiple users, was a primary design objective. The facility to add/remove users to and from on-going projects is impressive and, generally, works well. Unfortunately, SLRTOOL doesn't really allow users to collaborate, in any meaningful way. Due to this, much of its support for the SR process is quite limited.

A number of limitations are common across all (or most) of the candidate tools. Support for protocol development, by most tools, is generally quite limited. Only one tool, SLR-Tool, assisted this stage effectively. In addition, support for the search process (a frequently stressed issue within the community [5,6,7,8]), is largely absent. SLRTOOL is the only tool that provides an internal search facility (i.e. a facility within the tool for searching digital libraries). However, as noted in Section 4.2.3, its implementation is rather limited. Poor support for this aspect of an SR may be a consequence of the inherent difficulties associated with automated searching [7,19,20,21] which, we suggest need to be addressed, before effective tool support can be realised. Support for collaboration within a team-based SR is also limited. Only SLuRp provides reasonably effective facilities for collaboration amongst multiple users. It is believed that a deeper understanding of what are considered the *collaborative activities* in an SR is needed, in order to develop effective support.

5.2 Limitations of the Study

The main threats to validity arise from the subjective nature of many of the elements of the feature analysis process. The features used are essentially a preliminary set based on our own experiences and those reported in the literature. We hope that this exercise will provide the foundations for further study of the features expected from an SR tool. Similarly, the levels of importance, both for individual features and for feature sets, are

² Using the feature set weightings, as shown in Table 6.

based on experience. However, these can easily be adjusted and weighted scores re-calculated where priorities differ.

Of course the scoring is also subjective however as independent evaluators we have no vested interest in any of the candidate tools. Also, to mitigate any potential bias we performed a substantial validation exercise with all authors reviewing all scores for all tools.

The performance form feature (see Section 4.1.3) of SLuRp and ‘bulk-import’ feature of SLR-Tool (see Section 4.4.4) were not fully evaluated. It is, however, intended that these features be examined further, with additional support from their developers.

6. CONCLUSIONS

This study has evaluated four candidate tools; namely SLuRp, StArt, SLR-Tool and SLRTOOL, which aim to support the whole SR process. A set of features that such tools should possess has been developed and used as the criteria against which to evaluate the candidate tools.

SLuRp received the highest overall score and is, therefore, based on the results of this study; the most suitable tool to support SRs in SE. SLRTOOL received the lowest overall score, making it the least suitable.

The results of this study provide new insight into tools that support SRs in SE. We believe that one of its most interesting and significant outputs are the features presented in Table 1 and described in Section 3.2. The feature set is based on our assessment of what we believe an effective tool should include. The next stage is to circulate these features within the community in order to refine and validate them. It would also be interesting to explore SR tools in other domains to determine whether they could inform the development of tools in SE.

7. ACKNOWLEDGMENTS

The authors express thanks to the developers of each candidate tool for their cooperation. Thanks are also given to Keele University’s Environment, Physical Sciences and Applied Mathematics Research Institute for its partial support for Christopher Marshall.

8. REFERENCES

- [1] Kitchenham, B. A., Dyba, T., & Jorgensen, M. (2004). Evidence-based software engineering. ICSE 2004. *Proceedings. 26th International Conference on Software Engineering* pp. 273-281.
- [2] Kitchenham, B., Brereton, O. P., Budgen, D., Turner, M., Bailey, J., & Linkman, S. (2009). Systematic literature reviews in software engineering—a systematic literature review. *Information and software technology*, Vol. 51, no. 1, pp. 7-15.
- [3] Kitchenham, B. A., & Charters, S. (2007). Guidelines for performing systematic literature reviews in software engineering. *Keele University and University of Durham, EBSE Technical Report*
- [4] Staples, M., & Niazi, M. (2007). Experiences using systematic review guidelines. *Journal of Systems and Software*, 80(9), 1425-1437.
- [5] Riaz, M., Sulayman, M., Salleh, N., & Mendes, E. (2010). Experiences conducting systematic reviews from novices’ perspective. In *Proceedings of EASE* Vol. 10, pp. 1-10.
- [6] Babar, M. A., & Zhang, H. (2009). Systematic literature reviews in software engineering: Preliminary results from interviews with researchers. In *Proceedings of the 2009 3rd International Symposium on Empirical Software Engineering and Measurement*, pp. 346-355.
- [7] Brereton, P., Kitchenham, B. A., Budgen, D., Turner, M., & Khalil, M. (2007). Lessons from applying the systematic literature review process within the software engineering domain. *Journal of Systems and Software*, Vol. 80, no. 4, pp. 571-583.
- [8] Carver, J. C., Hassler, E., Hernandez, E., & Kraft, N. A. (2013). Identifying Barriers to the Systematic Literature Review Process. In *Empirical Software Engineering and Measurement, 2013 ACM/IEEE International Symposium on* (pp. 203-212). IEEE.
- [9] Marshall, C., & Brereton, P. (2013). Tools to Support Systematic Literature Reviews in Software Engineering: A Mapping Study. In *Empirical Software Engineering and Measurement, 2013 ACM/IEEE International Symposium on* (pp. 296-299).
- [10] Kitchenham, B., Linkman, S., & Law, D. (1997). DESMET: a methodology for evaluating software engineering methods and tools. *Computing & Control Engineering Journal*, 8(3), 120-126.
- [11] Kitchenham, B. A. (1997). Evaluating software engineering methods and tools, part 7: planning feature analysis evaluation. *ACM SIGSOFT Software Engineering Notes*, 22(4), 21-24.
- [12] Kitchenham, B. A., & Jones, L. (1997). Evaluating SW Eng. methods and tools, part 8: analysing a feature analysis evaluation. *ACM SIGSOFT Software Engineering Notes*, 22(5), 10-12.
- [13] Grimán, A., Pérez, M., Mendoza, L., & Losavio, F. (2006). Feature analysis for architectural evaluation methods. *Journal of Systems and Software*, 79(6), 871-888.
- [14] Hedberg, H., & Lappalainen, J. (2005). A preliminary evaluation of software inspection tools, with the DESMET method. In *Quality Software, 2005.(QSIC 2005). Fifth International Conference on* (pp. 45-52).
- [15] Bowes, D., Hall, T., & Beecham, S. (2012). SLuRp: a tool to help large complex systematic literature reviews deliver valid and rigorous results. In *Proceedings of the 2nd international workshop on Evidential assessment of software technologies*, pp. 33-36.
- [16] Hernandez, E., Zamboni, A., Fabbri, S., & Di Thommazo, A. (2012). Using GQM and TAM to evaluate StArt—a tool that supports Systematic Review. *CLEI Electronic Journal*, Vol. 15 no. 1, pp. 13-25.
- [17] M. Fernández-Sáez, M. G. Bocco, F.P. Romero. (2010). SLR-tool - a tool for performing systematic literature reviews. In *Proceedings of the 2010 International Conference on Software and Data Technologies*, pp. 144.
- [18] Kitchenham, B. A., Budgen, D., & Pearl Brereton, O. (2011). Using mapping studies as the basis for further research—a participant-observer case study. *Information and Software Technology*, 53(6), 638-651.
- [19] Dieste, O., Grimán, A., & Juristo, N. (2009). Developing search strategies for detecting relevant experiments. *Empirical Software Engineering*, 14(5), 513-539.
- [20] Bailey, J., Zhang, C., Budgen, D., Charters, S., & Turner, M. (2007). Search Engine Overlaps: Do they agree or disagree?. In *Realising Evidence-Based Software Engineering, 2007. REBSE'07. Second International Workshop on* (pp. 2-2). IEEE.
- [21] Dyba, T., Dingsoyr, T., & Hanssen, G. K. (2007). Applying systematic reviews to diverse study types: An experience report. In *Empirical Software Engineering and Measurement, 2007. ESEM 2007. First International Symposium on* (pp. 225-234). IEEE.